Mayank Singh

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Education

University of Arizona, College of Science — Tucson, AZ, USA
Ph.D. Candidate, Computer Science (Advisor: Prof. Eduardo Blanco) — Aug. 2023 – Present

New York University, Courant Institute of Mathematical Sciences (CIMS) — New York, NY, USA
M.S., Computer Science — Sept. 2014 – May 2016

Birla Institute of Technology and Science (BITS), Pilani – Goa Campus — Zuarinagar, GA, India
B.E. (Hons.), Computer Science — Aug. 2009 – Aug. 2013

Recent Work Experience

University of Arizona — Tucson, AZ, USA
Research Assistant, Department of Computer Science — Aug. 2023 – Present

- Conducting research on AI agents and agentic systems under Prof. Eduardo Blanco.

ServiceNow, Inc. — Kirkland, WA, USA
Senior Research Scientist Intern — Summers 2025 & 2026

- 2026: Conducted research within the Core AI team on approaches to improving the harnesses around LLMs in agentic systems; preprint forthcoming.
- 2025: Conducted research within the Core LLM team on better approaches to building multi-agent LLM systems, resulting in *Grammar Search for Multi-Agent Systems* (ACL 2026, Oral).

Maplebear, Inc. (Instacart) — San Francisco, CA, USA
Senior Machine Learning Engineer — Apr. 2023 – Aug. 2023

- Developed incentive optimization algorithms for shoppers during high-demand periods while working in the Availability & Coordination team.

Meta Platforms, Inc. (Facebook) — Menlo Park, CA, USA
Senior Software Engineer / Machine Learning Engineer — June 2016 – Jan. 2023

- Led the relevance and integrity areas in the Video Search Ranking team to improve the relevance of video search results to user queries and reduce the prevalence of inappropriate content.
- Created and improved product features for Facebook Search as part of the Content Search Product team.
- Worked on the Newsfeed Recommendations team to deliver better Reels recommendations to users.

Publications

- Mayank Singh, Vikas Yadav, and Eduardo Blanco. 2026.
  Error Taxonomy-Guided Prompt Optimization.
  Preprint. ArXiv
- Mayank Singh, Vikas Yadav, Shiva Krishna Reddy Malay, Shravan Nayak, Sai Rajeswar, Sathwik Tejaswi Madhusudhan, and Eduardo Blanco. 2026.
  Grammar Search for Multi-Agent Systems. In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, San Diego, USA. (Oral) Link
- Mayank Singh and Eduardo Blanco. 2024.
  Learning to Generate Rules for Realistic Few-Shot Relation Classification: An Encoder-Decoder Approach.
  In Findings of the Association for Computational Linguistics: EMNLP 2024, Association for Computational Linguistics, Miami, USA. Link
- Gajamohan Mohanarajah, Vladyslav Usenko, Mayank Singh, Raffaello D'Andrea, and Markus Waibel. 2015.
  Cloud-Based Collaborative 3D Mapping in Real-Time with Low-Cost Robots.
  IEEE Transactions on Automation Science and Engineering 12, 2 (2015), 423 – 431. Link

Additional Work Experience

Facebook, Inc. Menlo Park, CA, USA
Software Engineer Intern, Video Ads June 2015 – Aug. 2015

- Added support for creating Video Ads using Light-Weight Interfaces in Facebook Pages.

Cisco Systems (India) Private Limited Bengaluru, KA, India
Software Engineer I Aug. 2013 – Aug. 2014

- Contributed to implementing IP tunneling, key exchange, and cryptographic algorithms for the Cisco IOS and IOS-XR network operating systems.

ETH Zurich Zurich, ZH, Switzerland
Research Intern Jan. 2013 – June 2013

- Conducted Cloud Robotics research under Dr. Gajamohan Mohanarajah and Prof. Raffaello D'Andrea at the Institute of Dynamic Systems and Control.

Python Software Foundation Remote
Google Summer of Code Summers 2011 & 2012

- Extended hardware support in PySoy, a Linux game engine, by building a Wiimote IR pointer driver and a Bluetooth plugin for PlayStation 3 controllers.

Other Experience

- Recipient of the Tinker Research Grant from Thinking Machines Lab ($5,000 in compute credits, 2025).
- Conference/Journal Reviewing Experience:
  - NeurIPS 2026
  - ACL Rolling Review (Oct. 2024, Dec. 2024, May 2025, Oct. 2025, Jan. 2026)
  - COLING 2025
  - Journal of Computational Linguistics
- Teaching Assistant Experience:
  - University of Arizona: Introduction to Artificial Intelligence (Fall 2026); Algorithms (Spring 2026); Automata, Grammars and Languages (Fall 2025)
  - New York University: Deep Learning (Spring 2016); Financial Software Projects (Fall 2015); Music Software Projects (Fall 2015)
- Earned a certificate in Statistics and Data Science from the MITx MicroMasters Program in December 2020.
- Mentor for Google Summer of Code (2015) and Google Code-in (2012) with the CopyLeft Games Group.
- Represented New York University at the ACM ICPC Greater NY Regionals in 2014.

Useful Links

LinkedIn: linkedin.com/in/mayanks43/

GitHub: github.com/mayanks43/